CALDERWICK PUMP SYSTEMS LTD

# Major Incident Report INC-2025-014: Test Rig TR-3 Outage

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| **Incident ID** | INC-2025-014 |
| **Date of incident** | 14 August 2025 |
| **Site and asset** | Plant P2, Ardmoor; hydraulic test rig TR-3 |
| **Priority** | P1 |
| **Supplier involved** | Brennock Valve Services Ltd (VEN-014), contract CON-2024-031 |
| **Incident owner** | IT Service Manager |
| **Status** | Closed on 29 August 2025 |

## 1. Summary

On 14 August 2025 test rig TR-3 at the Ardmoor plant of Calderwick Pump Systems Ltd stopped during a customer endurance test. The service desk logged the incident as P1 and called the maintenance supplier. The fault was a failed PLC controller on the rig's hydraulic power unit. No spare controller was held on site, so the rig stayed out of service until a controller was taken from rig TR-1, which was idle that day. TR-3 was back in service the same evening. The customer test was repeated on 16 August 2025.

## 2. Timeline (14 August 2025)

| Time | Event |
|---|---|
| 06:40 | TR-3 stopped during the endurance test; the operator alerted the Test Rig Supervisor. |
| 06:55 | The service desk logged INC-2025-014 as P1 and called Brennock Valve Services Ltd. |
| 08:30 | Supplier engineer on site at TR-3. |
| 11:10 | Fault traced to a failed PLC controller on the hydraulic power unit; no bypass possible. |
| 13:00 | Plant Manager, Ardmoor approved taking the controller from idle rig TR-1. |
| 18:15 | Controller from TR-1 fitted and tested; TR-3 back in service. |

## 3. Measured times

Measured from the service desk log at 06:55: the supplier engineer arrived after 1 hour 35 minutes, and the rig was back in service after 11 hours 20 minutes.

## 4. Root cause

The controller failed after a firmware fault corrupted its configuration memory. The same firmware version runs on the controllers of TR-1, TR-2 and TR-4. The firmware was updated on all four rigs on 21 August 2025 by the OT Systems Engineer. A contributing cause was that Ardmoor held no spare controller.

## 5. Follow-up actions

| No. | Action | Owner (role) | Status |
|---|---|---|---|
| 1 | Buy two PLC controllers (one to replace the unit taken from TR-1, one spare to hold at Ardmoor). Purchase order PO-10234 was raised on 18 August 2025 to Ardent Controls Ltd (VEN-052) for USD 64,800 excluding tax, category OT hardware. | Procurement Manager, Plant Services | PO raised |
| 2 | Claim the service credit due from Brennock Valve Services Ltd for this incident under contract CON-2024-031. | Procurement Manager, Plant Services | Open |
| 3 | Update controller firmware on all four rigs. | OT Systems Engineer | Done 21 August 2025 |
| 4 | Add PLC controllers to the Ardmoor critical spares list. | Maintenance Lead, Ardmoor | Done 25 August 2025 |

## 6. Sign-off

Prepared by the IT Service Manager and reviewed by the Head of IT on 29 August 2025.